Corporate Finance Institute®

Data Visualization & Dashboards - The Basics

Term	Definition
Actionable Insights	Actionable insights are conclusions drawn from data analysis that can be implemented to drive decision-making and improve outcomes. These insights provide clear recommendations and focus on specific actions that can be taken to achieve desired results.
Analytical Dashboard	An analytical dashboard is a data visualization tool that provides users with deep insights into key performance metrics through interactive elements and detailed visual representations. It is designed for in-depth analysis and decision-making, often allowing users to drill down into data for more granular insights.
Annotations	Annotations are notes or labels added to visual elements in a data visualization to provide context, explanations, or highlight important information. They enhance understanding and help guide the viewer's interpretation of the data.
Area Chart	An area chart is a graphical representation that displays quantitative data visually by filling the area between a line and a baseline, effectively showing the cumulative total over time. It is useful for illustrating trends and comparing the contribution of different categories to the whole.
Audience	**Audience**: In the context of data visualization and dashboards, the audience refers to the specific group of people who will be viewing and interpreting the visualized data. Understanding the audience's needs, preferences, and expertise is crucial for designing effective visualizations that communicate insights clearly and meaningfully.
Bar Chart	A bar chart is a visual representation of data that uses rectangular bars to compare different categories or groups. The length of

	each bar corresponds to the value it represents, making it easy to see variations between different items.
Bullet Chart	A bullet chart is a visual representation that displays a single measure against a qualitative range and a target, typically represented by a horizontal bar. It effectively conveys performance data and comparisons within a compact space, making it ideal for dashboard applications.
Chart Design	Chart design refers to the process of creating visual representations of data in a clear and effective manner. It involves selecting appropriate chart types, colors, and layouts to enhance comprehension and facilitate decision-making.
Clarity	Clarity in data visualization refers to the ease with which viewers can understand the information being presented. It involves using clear labels, well-defined visuals, and organized layouts to effectively convey the intended message without confusion.
Cognitive Load	Cognitive load refers to the amount of mental effort and processing capacity required to understand information. In data visualization, managing cognitive load is essential to enhance comprehension and facilitate effective decision-making by minimizing unnecessary complexity.
Cohesive Stories	Cohesive stories in data visualization refer to narratives that effectively integrate data elements to convey a clear, unified message or insight. They guide the audience through a logical progression, making complex information accessible and engaging.
Color Scheme	A color scheme refers to the planned selection of colors used in data visualizations to enhance clarity and impact. It helps convey information effectively and ensures that visual elements are distinguishable and aesthetically pleasing.
Correlation Matrix	A correlation matrix is a table that displays the correlation coefficients between multiple variables, helping to identify relationships and patterns among them. It visually represents the strength and direction of these correlations, making it easier to understand how different variables are interrelated.
Data Quality	Data Quality refers to the accuracy, completeness, and reliability of data used in analysis and reporting. High-quality data is

	essential for making informed decisions and generating trustworthy insights in data visualization and dashboards.
Data Story	A data story is a narrative that combines data, visualizations, and context to convey insights effectively. It helps audiences understand complex information by presenting it in an engaging and relatable way.
Exploratory Dashboard	An exploratory dashboard is an interactive tool designed to help users analyze and understand data by allowing them to explore various dimensions and metrics. Unlike traditional dashboards, which focus on monitoring key performance indicators, exploratory dashboards emphasize data discovery and insight generation through user-driven exploration.
Interactivity	Interactivity refers to the ability of users to engage with data visualizations and dashboards in real-time, allowing them to manipulate data views and gain insights through exploration. This dynamic feature enhances user experience by providing personalized data analysis tailored to individual needs.
Key Metrics	Key metrics are essential measurements that help evaluate the performance and success of a business, project, or initiative. They provide critical insights that guide decision-making and strategic planning.
Narrative	A narrative in data visualization refers to the storytelling element that guides viewers through the data presented. It helps to contextualize the information, highlighting key insights and trends to facilitate understanding and engagement.
Performance	Performance refers to the effectiveness and efficiency with which tasks and goals are achieved within a given context. In data visualization, it often relates to how well visual representations communicate insights and facilitate decision-making.
Pie Chart	A pie chart is a circular statistical graphic that is divided into slices to illustrate numerical proportions. Each slice represents a category's contribution to the whole, making it easy to compare relative sizes at a glance.
Scatter Plot	A scatter plot is a graphical representation that displays two variables as points in a Cartesian coordinate system. It helps

	identify relationships, trends, or correlations between the variables by showing how they vary together.
Simplicity	Simplicity in data visualization refers to the practice of presenting information in a clear and straightforward manner, minimizing unnecessary complexity. By focusing on essential elements and avoiding clutter, simplicity enhances the viewer's understanding and interpretation of the data.
Stakeholders	Stakeholders are individuals or groups that have an interest in or are affected by a project or its outcomes. In the context of data visualization and dashboards, stakeholders can include decision-makers, team members, and end-users who rely on insights to drive actions and strategies.
Transparency	Transparency in data visualization refers to the clarity and openness of how data is presented, allowing viewers to easily understand the information being conveyed. It enables audiences to trust the visualizations by ensuring that data sources, methods, and potential biases are clearly disclosed.
Trend	A trend is a general direction in which data points are moving over time, indicating a pattern or tendency. Analyzing trends helps identify changes in behavior, performance, or outcomes, informing decision-making and strategy.
Visual Attributes	Visual attributes are the characteristics of graphical elements used in data visualization, including color, shape, size, and texture. These attributes help convey information effectively and enhance the viewer's understanding of the data presented.
Visualization	Visualization is the graphical representation of data or information, allowing for easier interpretation and understanding. It transforms complex datasets into visual formats, such as charts and graphs, to highlight trends, patterns, and insights.
Waterfall Chart	A waterfall chart is a data visualization tool used to illustrate the cumulative effect of sequentially introduced positive or negative values. It helps to visualize the incremental changes in a data series, making it easier to understand how an initial value is influenced over a period of time or through various factors.
X Axis	The X Axis is the horizontal line on a graph, typically representing the independent variable. It helps to organize and display data

	points in relation to one another, allowing for easy comparison and analysis.
Y Axis	The Y Axis is the vertical line on a graph that represents the scale of values or measurements. It is commonly used to display quantitative data, allowing for the visualization of trends and comparisons with respect to the X Axis.